

ASSESSMENT (3)

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1) Footprinting tools Conducting Competitive Intelligence.

In a passive security assessment, the consultant collects information from publicly available sources without directly connecting to a proving ABC Technologies systems. used as follows.

(i) Domain Registration Details - 'WHOIS'

Tools: WHOIS, ICANN Lookup.

It can provide publicly available domain Registration such as:

- Domain Name and Registration Date
- Register Information
- Name Servers
- Domain Status

(ii) Employee Email Address the Harvester

The harvester gathers publicly available information from Search Engines, websites, other sources.

- Employee Names
- Email Address
- Domain Names
- publicly Indexed hosts

(iii) Technology Stack - wappalizer/ what web

These tools can identify technology used by publicly accessible website

- web servers
- Programming languages
- CMS platforms.

(iv) Publicly Exposed Subdomains

Amass can perform passive Reconnaissance using publicly available datasets and sources to discover subdomains discovered with domains

Ex :

- mail. ex. com
- blog. ex. com
- dev. ex. com

2. DNS Zone Transfers.

A DNS Zone Transfer is a mechanism used to replicate DNS records between authoritative DNS servers. During an authorized penetration test, the analyst can check whether the DNS server incorrectly permits zone transfer.

(i) Identify the Authoritative DNS Servers.

First, determine the target's Name Servers

`nslookup -type=NS example.com`

This identifies authoritative DNS Servers

(ii) Test for Zone Transfers.

The analyst can use the dig Command:

Here :

- AXFR Requests a Complete DNS Zone.

Transfer

3. Analyze Retrieved Records.

If the Zone Transfer exceeds, the Return Records Reveal:

- Internal or Infrastructure hosts
- Hostname and IP address.
- Developing or Testing Systems
- VPN-Related hosts

Mail Servers:

Sub domains:

This Information helps the security analyst understand the organization's DNS Structure and Identify Systems that Require further Security Testing

3. Introduction to Social Engineering Social Engineering attacks and Countermeasures;

The email described is a phishing attack specifically an email based credential phishing attack. The attacker pretends to be from the organization IT department and creates a sense of urgency to trick employees a fake website.

- (i) Identify the sender
- (ii) Do not click the link immediately
- (iii) Verify through an official portal
- (iv) Check website address
- (v) Never share password, OTPs
- (vi) Report the suspicious Email
- (vii) use Multi-factor Authentication.

The attack is phishing, using urgency and impersonation to steal employee credentials. By verifying the sender, avoiding suspicious links, independently contacting IT, checking URLs, protecting password and OTPs, employees can greatly reduce the risk of credential theft.